

PANDA Component Description: *rs232_simple*

Overview

The *rs232_simple* component implements a simple RS-232 port. It does not contain any input or output buffers. The baudrate is defined through a generic. The component has the ability to generate interrupts if a databyte is sent or received.

Generics

Name	Type	Default	Description
clk_frequency	integer	10'000'000	Clock frequency in Hz
baudrate	integer	9'600	Baudrate in Baud

External Ports

Name	Type	Width	Description
rx	std_logic	1	Serial input data
tx	std_logic	1	Serial output data

Registers

Data Register

Name: DAT
Address: Base + 0x00

Name	DAT
R/W	R/W
Reset Value	0x00

The *DAT* register represents the received data and data to send. After a write operation on the *DAT* register, the written data is sent over RS-232. A read operation on the *DAT* register returns the last byte received over RS-232.

A write operation on *DAT* clears the *SND* bit in *INT*, a read operation on *DAT* clears the *REC* bit in *INT*.

If *DAT* is written while data is sent, the write operation does have no effect. The written data will not be sent.

Configuration Register

Name: CFG

Address: Base + 0x01

Name	-	-	-	-	-	BUSY	REC	SND
R/W	-	-	-	-	-	R	R/W	R/W
Reset Value	0	0	0	0	0	0	0	0

SND: '1' → Enable interrupt if data is sent (interrupt if *SND* in *INT* is set)
 '0' → Disable interrupt if data is sent

REC: '1' → Enable interrupt on receiving data (interrupt if *REC* in *INT* is set)
 '0' → Disable interrupt on receiving data

BUSY: '1' → The component is sending data over the *tx* line at the moment
 '0' → No data is sent at the moment, ready for new data

Interrupt Register

Name: INT

Address: Base + 0x02

Name	-	-	-	-	-	REC	SND
R/W	-	-	-	-	-	R/W	R/W
Reset Value	0	0	0	0	0	0	0

SND: '1' → Data send over RS-232 completed since last clear
 '0' → No data send over RS-232 completed since last clear

REC: '1' → Received data over RS-232 since last clear
 '0' → Received no data over RS-232 since last clear

Both bits can be cleared by writing a '0' to them. Writing a '1' to these bits does not affect their state. They can only be cleared but not set by software.

The *REC* bit is also cleared if *DAT* is read, the *SND* bit is cleared if *DAT* is written.

Unused Register

Address: Base + 0x03

This address is not used. Write operations on it do not have any effect. Read operations result in a random value.

Driver Macros

WAIT_SND

Call: *_WAIT_SND [data]

Parameter	Description	Immediate/Variable	Size
data	Data to send	Immediate/Variable	8-Bit

Waits until the comport is not busy and then sends [data] over the comport.

WAIT_REC

Call: *_WAIT_REC [variable]

Parameter	Description	Immediate/Variable	Size
variable	Variable to save the received data in	Variable	8-Bit

Waits until data is received and then saves the data into [variable]. If a received databyte is not read when the macro is called, it returns this byte immediately.